

Theoretical Concepts of One Hot Encoding

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OHE(One Hot Encoding) is used to Convert Nominal Categorical Data into Numerical Data.

→ No Natural Order in the Data

For Example:

Color	Target
Yellow	0
Yellow	1
Blue	1
Yellow	1
Red	1
Yellow	0
Red	1
Red	0
Yellow	1
Blue	0

Color_Y	Color_B	Color_R	Target
1	0	0	0
1	0	0	1
0	1	0	1
1	0	0	1
0	0	1	1
1	0	0	0
0	0	1	1
0	0	1	0
1	0	0	1
0	1	0	0

One Hot Encoding

$[Y, B, R]$
0 1 2

From this We can Understand:

1. Distinct Columns For each color(category).
2. We can represent like vector. E.g: Yellow = $[1, 0, 0]$
3. If It has 50/60 categories we have to make 50/60 Columns.

Dummy Variable Trap :

• Multicollinearity:

If you have n columns after one-hot encoding, you should remove one column, keeping only $n - 1$. Usually, we remove the first column.

• Why do we remove a column?

Let's say you have 3 columns representing a category (e.g., Color: Red, Blue, Yellow). You check for any mathematical relationships between them. If there is a dependency (like one column can be predicted from the others), that's a problem.

In machine learning, especially in linear and logistic regression, we assume that all input columns should be independent — they should not have relationships with each other.

• Take this example:

Suppose you one-hot encode 3 categories — [Red, Blue, Yellow]. Now, each row has a combination like $[1, 0, 0]$ or $[0, 1, 0]$. But here's the catch: if you know any two values, you can always predict the third. For example, if $[Red = 0, Blue = 1]$, then Yellow must be 0. This kind of dependency causes multicollinearity.

This is why we drop one dummy column (usually the first one) — to break this mathematical relationship and avoid model issues.

OHE using most frequent variables

Let's See an Example:

	brand	km_driven	fuel	owner	selling_price
0	Maruti	145500	Diesel	First Owner	450000
1	Skoda	120000	Diesel	Second Owner	370000
2	Honda	140000	Petrol	Third Owner	158000
3	Hyundai	127000	Diesel	First Owner	225000
4	Maruti	120000	Petrol	First Owner	130000

Now let's focus on the **brand** of the cars.

There are around **40 to 50 different car brands**, maybe even more. If we check:

```
⇒ df['brand'].value_counts()
```

we'll see that some car brands appear **very frequently** — like 100, 200, or 250 times. But others are very **rare**, like:

- Rolls Royce
- Range Rovers
- or other expensive cars.

These rare brands don't show up often, but they can still affect the model if each is turned into a separate column by One Hot Encoding.

So What Can We Do?

We can **group these rare brands** into one **"Other"** category or a single column. This way:

- We **reduce the number of columns** created by One Hot Encoding.
- We keep the dataset **less sparse and more efficient**.
- We **prevent overfitting** on rare categories.

This is a great way to **reduce dimensionality** while keeping the important patterns.

Summary:

- One Hot Encoding turns each unique category into a new column.
- Too many unique values (like 50 brands) means **too many columns**.
- Solution: group less frequent ones into a new column like "other_brand".